Strategy research (research period 2012-2021 only)

1. The grid I tried (every variant, nothing hidden)

k	hold	side	gross Sharpe	net Sharpe	Sharpe 12-16	Sharpe 17-21	CAGR %	MaxDD %	trades	time in market %
0.5	1	both	1.62	0.97	1.16	0.82	13.05	-14.95	860	54.91
0.5	1	long	1.47	0.96	1.19	0.78	10.58	-18.72	563	39.73
0.5	2	both	1.68	1.25	1.21	1.28	20.19	-16.74	661	77.7
0.5	2	long	1.47	1.17	1.19	1.16	15.76	-19.67	385	57.16
0.5	3	both	1.44	1.09	1.11	1.07	18.13	-23.32	578	88.13
0.5	3	long	1.23	0.99	0.97	1.02	13.85	-20.18	318	64.98
1	1	both	1.58	1.07	1.21	0.98	10.93	-15.48	511	26.35
1	1	long	1.52	1.08	1.03	1.13	8.48	-8.8	340	18.29
1	2	both	1.47	1.13	1.01	1.24	14.82	-9.55	430	43.78
1	2	long	1.44	1.16	1.1	1.22	11.71	-8.93	273	30.69
1	3	both	1.17	0.9	0.94	0.87	12.53	-14.07	381	55.77
1	3	long	1.2	0.99	1.21	0.83	11.04	-10.43	233	39.4
1.5	1	both	1.01	0.66	0.74	0.62	4.93	-13.71	269	12.36
1.5	1	long	1.04	0.76	0.57	0.91	4.43	-6.78	167	7.82
1.5	2	both	1.23	1	0.98	1.03	10.5	-12.91	244	21.89
1.5	2	long	1.22	1.03	1.19	0.9	7.63	-9.72	147	14.12
1.5	3	both	1.07	0.88	1.08	0.75	9.92	-16.08	223	29.87
1.5	3	long	1.06	0.9	1.16	0.72	7.52	-11.39	132	19.31

Variants passing the rule (>= 100 trades, positive net Sharpe in both halves): 18 of 18
Chosen: k=0.5, hold=2 day(s), side=both, window=60

2. Did I just pick the luckiest of 18 variants? (White's Reality Check)

Bootstrap p-value that the best of the 18 variants is no better than zero-edge luck: **0.003**
(small = the best variant is hard to explain by selection luck alone)

3. Chosen rule, research period

	Strategy (net of costs)	Strategy 2012-16	Strategy 2017-21	Buy & hold Nifty (open to open)
CAGR %	20.19	17.89	22.5	14.4
Ann vol %	15.76	14.43	16.98	17.64
Sharpe	1.25	1.21	1.28	0.85
Max drawdown %	-16.74	-16.74	-11.57	-37.77
Days	2444	1216	1228	2444
Trades	661	333	328	nan
Win rate %	51.89	51.05	52.74	nan
Avg win %	1.55	1.53	1.57	nan
Avg loss %	-1.07	-1.08	-1.07	nan

Gross Sharpe (before costs): 1.68. Total costs paid over the period: 66.1% of capital (simple sum of daily costs).
Time in the market: 78% of days. Average trade: 0.29% net.

Net return by calendar year (%):

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
net return %	22.5	6.6	28.9	8.9	20.7	2.4	0.7	25.7	68.5	23.1

Gross P&L split: long side 189.0% of capital, short side 67.5% of capital (sum of daily gross, no compounding).
Net Sharpe excluding 2020: 1.13 (full period 1.25)

Same rule, same parameters on **Bank Nifty** (no re-tuning): net Sharpe 0.79, CAGR 16.7%, max DD -26.7%, 687 trades.

4. How much do costs matter?

cost per side %	net Sharpe	CAGR %
0	1.68	28.65
0.02	1.46	24.35
0.05	1.25	20.19
0.08	1.03	16.16
0.1	0.81	12.27

5. Parameter sensitivity (each parameter moved by -20% / +20%)

Hold days is an integer so I moved it by one day (h-1, h+1); with a 1-3 day hold, 20% is less than a day. The window is rounded to whole days.

variant	k	window	hold	net Sharpe	CAGR %	MaxDD %	trades
base	0.5	60	2	1.25	20.19	-16.74	661
k -20%	0.4	60	2	0.99	16.01	-19.38	716
k +20%	0.6	60	2	1.23	19.34	-18.6	615
window -20%	0.5	48	2	1.15	18.56	-17.27	672
window +20%	0.5	72	2	1.32	21.67	-16.27	654
hold -1	0.5	60	1	0.97	13.05	-14.95	860
hold +1	0.5	60	3	1.09	18.13	-23.32	578
corner k-20% w-20% h-1	0.4	48	1	0.73	10.04	-16.47	936
corner k-20% w-20% h+1	0.4	48	3	0.91	15.16	-25.34	653
corner k-20% w+20% h-1	0.4	72	1	0.87	12.17	-13.65	909
corner k-20% w+20% h+1	0.4	72	3	1	16.62	-25.24	637
corner k+20% w-20% h-1	0.6	48	1	0.95	12.34	-14.47	787
corner k+20% w-20% h+1	0.6	48	3	1.05	17.23	-22.98	537
corner k+20% w+20% h-1	0.6	72	1	0.96	12.41	-15.48	755
corner k+20% w+20% h+1	0.6	72	3	1.01	16.26	-26.42	521

Net Sharpe is positive in 100% of these 15 variants. Range: 0.73 to 1.32.

6. Drawdown I should realistically expect (stationary bootstrap, 10,000 paths)

Over a 1000-day (~4 year) horizon, resampling this strategy's own daily net returns.

Scenario 1: the edge is exactly what I measured in the research period

	max drawdown %	annualised return %	Sharpe
typical (median)	-14.9	20	1.3
bad (worst 25%)	-18.1	14.4	1
worse (worst 10%)	-21.9	9.5	0.7
very bad (worst 5%)	-24.6	6.7	0.5
extreme (worst 1%)	-29.6	1.9	0.2

P(4-year max drawdown worse than 10%) = 96%, worse than 15% = 50%, worse than 20% = 16%, worse than 25% = 4%. P(4-year Sharpe <= 0) = 0.2%.

Scenario 2: only half of the measured edge is real (daily mean return cut by 50%)

	max drawdown %	annualised return %	Sharpe
typical (median)	-19.8	8.8	0.6
bad (worst 25%)	-24.8	3.7	0.3
worse (worst 10%)	-30.6	-0.7	0
very bad (worst 5%)	-34.2	-3.3	-0.2
extreme (worst 1%)	-40.9	-7.6	-0.5

P(4-year max drawdown worse than 10%) = 99%, worse than 15% = 81%, worse than 20% = 49%, worse than 25% = 24%. P(4-year Sharpe <= 0) = 9.2%.

Historical max drawdown of the rule in the research period: -16.7%.

Caveat: the bootstrap only captures sampling noise. It cannot see regime change or the selection bias of picking this rule out of many ideas, so scenario 2 is the one i would plan around.

7. How much is just market exposure?

Average net exposure 0.37 of capital. Beta to always-long Nifty 0.33 (t=4.6). Alpha 14.7% a year (t=3.02).